

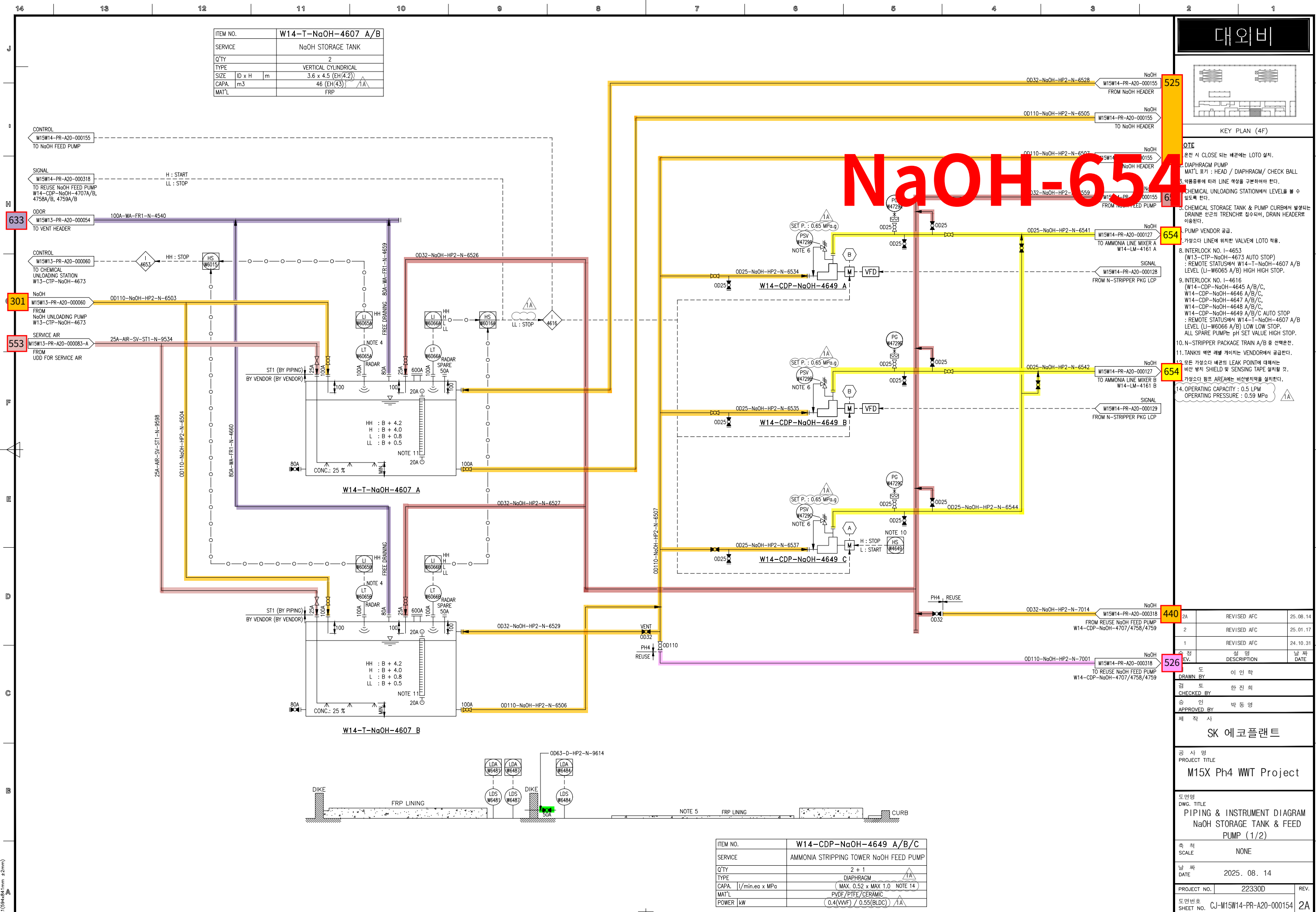
DRAWING LIST FOR TEST PACKAGE

NAOH-654

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	654	4FA	OD25-NaOH-HP2-N-6541-01	1 OF 2	0	CJ-M15W14-PR-A20-000154		
2	654	4FA	OD25-NaOH-HP2-N-6541-02	2 OF 2	0	CJ-M15W14-PR-A20-000154, 127		
3	654	4FA	OD25-NaOH-HP2-N-6542-01	1 OF 2	0	CJ-M15W14-PR-A20-000154		
4	654	4FA	OD25-NaOH-HP2-N-6542-02	2 OF 2	0	CJ-M15W14-PR-A20-000154, 127		
5	654	4FA	OD25-NaOH-HP2-N-6544-01	1 OF 1	0	CJ-M15W14-PR-A20-000154		
6	654	5FA	OD25-NaOH-HP2-N-6541-01	1 OF 1	0	CJ-M15W14-PR-A20-000127		
7	654	5FA	OD25-NaOH-HP2-N-6542-01	1 OF 1	0	CJ-M15W14-PR-A20-000127		
8	654	RFA	OD25-NaOH-HP2-E-6541-01	1 OF 1	0	CJ-M15W14-PR-A20-000127		
9	654	RFA	OD25-NaOH-HP2-E-6542-01	1 OF 1	0	CJ-M15W14-PR-A20-000127		
10	654	4FA	OD32-NaOH-HP2-N-6559-02	2 OF 2	0	CJ-M15W14-PR-A20-000154		
11								
12								



ITEM NO.	W14-T-NaOH-4607 A/B
SERVICE	NaOH STORAGE TANK
Q'TY	2
TYPE	VERTICAL CYLINDRICAL
SIZE	ID x H m 3.6 x 4.5 (EH(4.2))
CAPA.	m3 46 (EH(43))
MAT'L	FRP

ITEM NO.	W14-CDP-NaOH-4649 A/B/C
SERVICE	AMMONIA STRIPPING TOWER NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	l/min.ea x MPa (MAX. 0.52 x MAX 1.0 NOTE 14)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.4(VWF) / 0.55(BLDC))

대외비

KEY PLAN (4F)

OTE
혼전 시 CLOSE 되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL
3. 악물종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAINE 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이동한다.
6. PUMP VENDOR 공급.
7. 가성소다 LINE에 위치한 VALVE에 LOTO 적용.
8. INTERLOCK NO. I-4653
(W13-CTP-NaOH-4673 AUTO STOP)
: REMOTE STATUS에서 W14-T-NaOH-4607 A/B LEVEL (U-W6065 A/B) HIGH HIGH STOP.
9. INTERLOCK NO. I-4616
(W14-CDP-NaOH-4645 A/B/C,
W14-CDP-NaOH-4646 A/B/C,
W14-CDP-NaOH-4647 A/B/C,
W14-CDP-NaOH-4648 A/B/C,
W14-CDP-NaOH-4649 A/B/C AUTO STOP
: REMOTE STATUS에서 W14-T-NaOH-4607 A/B LEVEL (LI-W6066 A/B) LOW LOW STOP.
ALL SPARE PUMP는 pH SET VALUE HIGH STOP.
10. N-STRIPPER PACKAGE TRAIN A/B 중 선택운전.
11. TANK의 액면 레벨 경계치는 VENDOR에서 공급한다.
12. 모든 가성소다 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
가성소다 환부 AREA에는 비산방지막을 설치한다.
14. OPERATING CAPACITY : 0.5 LPM
OPERATING PRESSURE : 0.59 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 EV.	실명 DESCRIPTION	날짜 DATE

도인학

이인학

한진희

박동명

제작사

SK 에코플랜트

공사명

PROJECT TITLE

M15X Ph4 WWT Project

도면명

DWG. TITLE

PIPING & INSTRUMENT DIAGRAM

NaOH STORAGE TANK & FEED PUMP (1/2)

축척

SCALE

NONE

날짜

DATE

2025. 08. 14

PROJECT NO.

22330D

REV.

A

도면번호

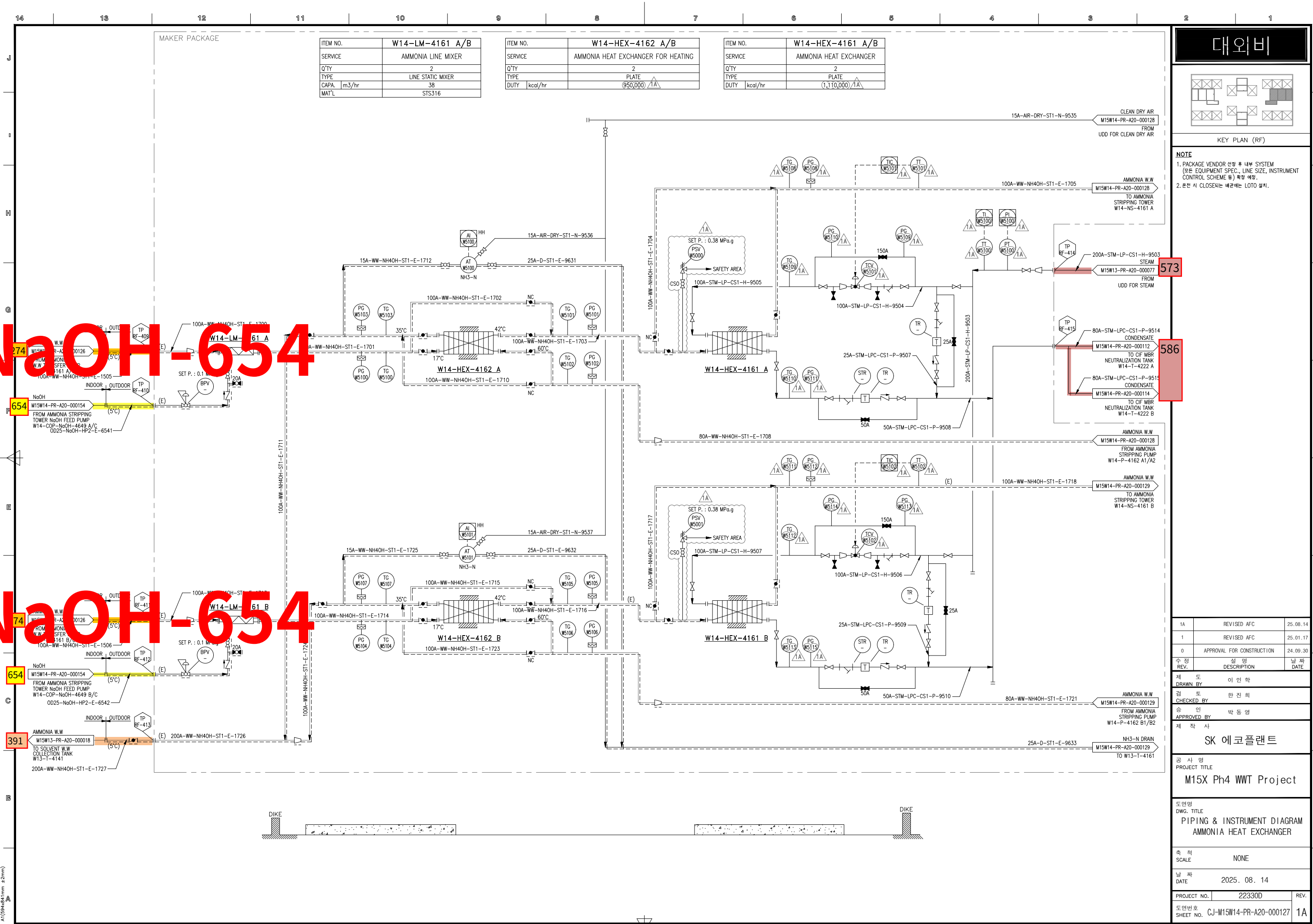
SHEET NO.

CJ-M15W14-PR-A20-000154

2A

NaOH-654

NaOH-654



대외비

KEY PLAN (RF)

NOTE

1. PACKAGE VENDOR 선정 후 내부 SYSTEM (또는 EQUIPMENT SPEC., LINE SIZE, INSTRUMENT CONTROL SCHEME 등) 확정 예정.

2. 로트 시 CLOSE되는 배관에는 LOTO 설치.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE
제 도	이 인 학	
DRAWN BY		
검 토	한 진 회	
CHECKED BY		
승 인	박 동 명	
APPROVED BY		
제 작 사	SK 에코플랜트	
공 사 명	PROJECT TITLE	
M15X Ph4 WWT Project		
도면명	DWG. TITLE	
PIPING & INSTRUMENT DIAGRAM AMMONIA HEAT EXCHANGER		
속 척	SCALE	
NONE		
날 짜	DATE	
2025. 08. 14		
PROJECT NO.	22330D	REV.
도면번호	SHEET NO. CJ-M15W14-PR-A20-000127	
SHEET NO.	1A	



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB2/VB2L0	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

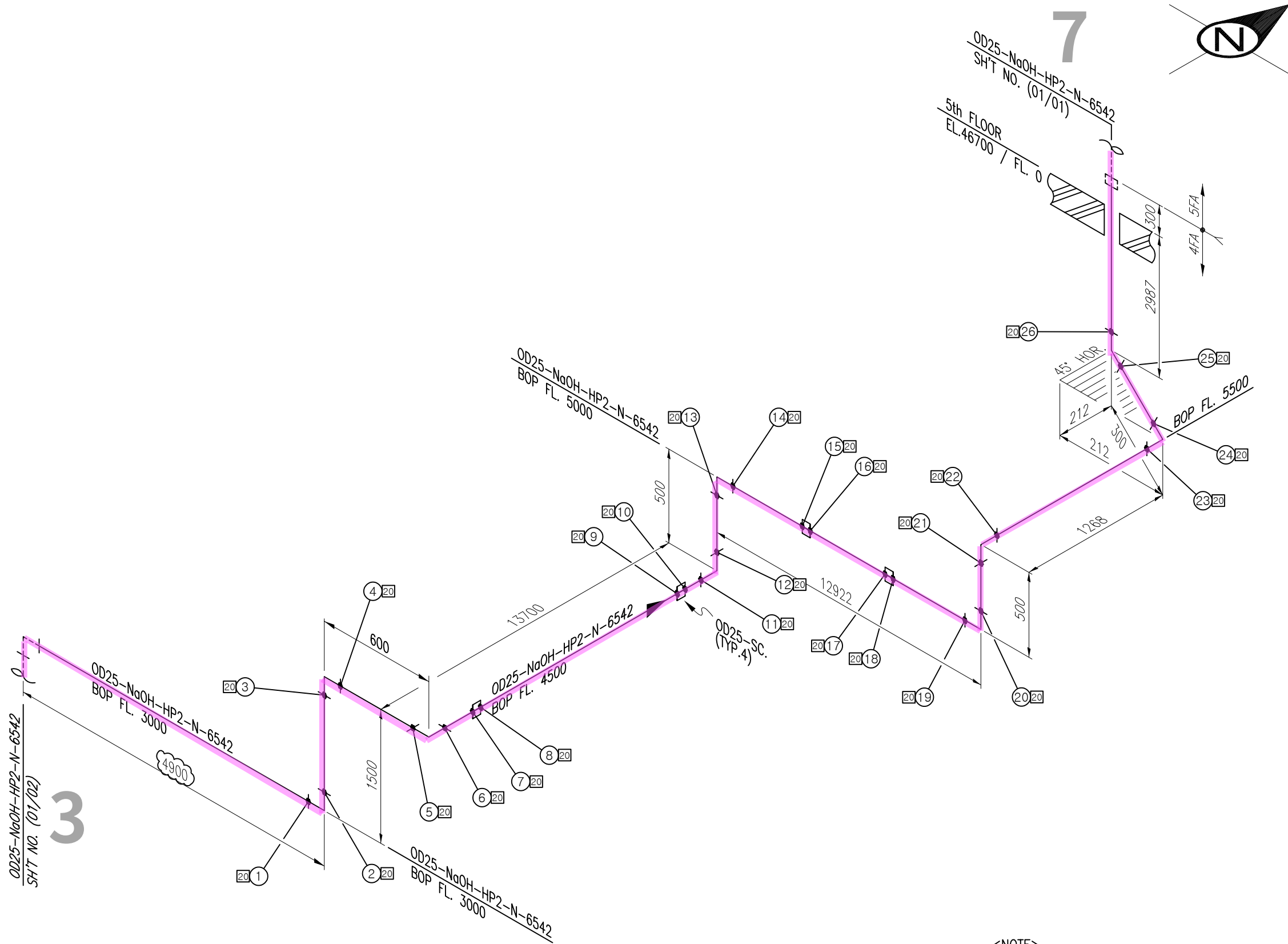
정고: 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유보자의 불법 사용으로 피해가 발생할 경우 유보자에게 민/형사상 책임은 요구할 수 있음



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB2/VB2LO	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	—	DESIGN	6.8	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040009-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY NaOH	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000154		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. OD25-NaOH-HP2-N-6541	SH'T NO.	REV.
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		02 / 02	1

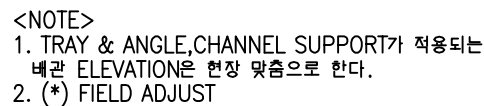
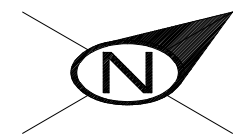


<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

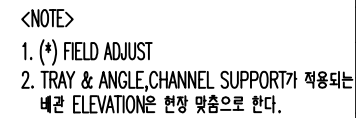
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR 	PROJECT NAME	
	THICK.	—	DESIGN	6.8	60	UNIT	4FA									M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ—M15W14—PI—A30—040009—A			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	NaOH	JOB NO.	
						REF. P & ID CJ—M15W14—PR—A20—000154			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD25—NaOH—HP2—N—6542	SH'T NO. 02 / 02	REV. 1
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				



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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	6.8	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040012-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOH	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000154								OD25-NaOH-HP2-N-6544	01 / 01	0

정고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
NaOH		22330D	
LINE NO.	SH'T NO.	REV.	
OD25-NaOH-HP2-N-6541	01 / 01	0	

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	6.7	60	UNIT	5FA								M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-050008-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOH	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000154								OD25-NaOH-HP2-N-6541	01 / 01	0

※고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있으며



FULL COUPLING : FC * HALF COUPLING : HF/HF1
VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
SPN * EMERGENCY SHOWER : SSH
H * SPECIAL BOLT & NUT : BL/BL1

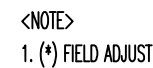
CLIENT  8

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



9

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY NaOH		JOB NO. 22330D	
LINE NO.	0D25-NaOH-HP2-E-6542	SH'T NO. 01 / 01	REV. 1



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHIELDED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	25	DESIGN	6.8	60	UNIT RFA								M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10005-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY NaOH	JOB NO. 22330D	
						REF. P & ID CJ-M15W14-PR-A20-000127		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. OD25-NaOH-HP2-E-6542	SH'T NO. 01 / 01	REV. 1
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

